Oct 27, 2026 (Due: 08:00 Nov 3, 2026)

1. Write a program to compute all eigenvectors of an upper triangular matrix with distinct diagonal entries. Compare your result with the one produced by a math library (e.g., `eig` in MATLAB/Octave).

2. Implement the tridiagonal reduction algorithm for Hermitian matrices. Explain why we can require subdiagonal entries to be real and nonnegative.

(optional) Test your implementation with a few A's and $A + \Delta A$'s. How do the unitary transformation and the tridiagonal matrix change according to the perturbations?

3. Let $D \in \mathbb{R}^{n\times n}$ be diagonal with distinct eigenvalues, $z \in \mathbb{C}^n$ be a vector with no zero entries, and $\rho \in \mathbb{R} \setminus \{0\}$. Suppose that (λ, u) is an eigenpair of $D + \rho zz^*$. Show that
(1) $\lambda I - D$ is nonsingular;
(2) $(\lambda I - D)^{-1}z$ is an eigenvector of $D + \rho zz^*$;
(3) $z^*u \neq 0$.

4. Randomly generate a relatively small (e.g., 6×6) real symmetric matrix of the form $A = \operatorname{diag}\{d_1, d_2, \ldots, d_n\} + zz^\top$, where d_i's are distinct, and z does not have any zero entry. Visualize the function

$$f(\lambda) = 1 - \sum_{i=1}^{n} \frac{z_i^2}{\lambda - d_i}.$$

Highlight the eigenvalues of A in the plot and make sure they match the roots of $f(\lambda)$. For simplicity, you may compute the eigenvalues of A by existing functions from math libraries (e.g., `eig` from MATLAB/Octave).

5. Implement the Jacobi diagonalization algorithm for real symmetric matrices. Visualize the performance for a few matrices with different sizes. Visualize the convergence history for one example. You are encouraged to make observations on *componentwise* convergence.

You are also encouraged to try the "wrong" choice of Jacobi rotations for the cyclic Jacobi algorithm.

6. (H) Implement the cyclic reduction algorithm for solving tridiagonal linear systems. Test your implementation with the examples in Exercise 5 from the homework on Sep. 22.

(optional) How to solve banded linear systems using cyclic reduction?

7. (optional) Design an algorithm to perform diagonal swapping of the real Schur form

$$A = \begin{bmatrix} A_{1,1} & A_{1,2} \\ & A_{2,2} \end{bmatrix},$$

where $A_{1,1}$, $A_{2,2} \in \mathbb{R}^{2\times 2}$.

8. (optional) Let $D = \operatorname{diag}\{d_1, \ldots, d_n\}$ be a real diagonal matrix. Let α_1, …, α_n be real scalars that satisfy

$$d_n < \alpha_n < \cdots < d_i < \alpha_i < d_{i-1} < \alpha_{i-1} < \cdots < d_1 < \alpha_1.$$

Show that the α_i's are the exact eigenvalues of $D + uu^\top$, where entries of the real vector u are defined by

$$u_i = \left(\frac{\prod_{1\le j\le n}(\alpha_j - d_i)}{\prod_{1\le j\le n,\, j\neq i}(d_j - d_i)} \right)^{1/2}, \qquad (1 \le i \le n).$$

9. (H, optional) Describe how to perform aggressive early deflation on a real symmetric tridiagonal matrix. Implement your algorithm and test it with a few examples.